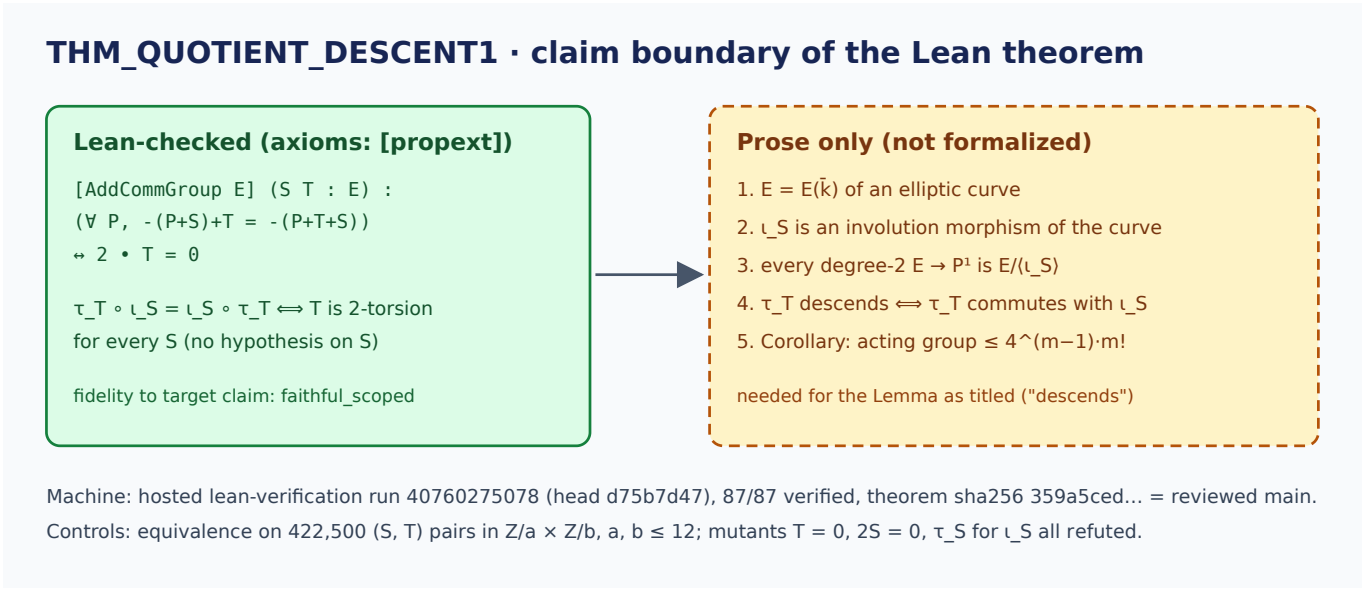


Quotient translation-descent: independent semantic review — 2026-10-29

Decision (scoped): `CryptoResearch.Quotient.translation_commutes_iff_two_torsion`
is a **faithful_scoped** formalization of the claim in
`formal/targets/quotient-translation-descent.yaml`: the group-law identity
"τ_T commutes with ι_S for every S exactly when 2T = 0". It does **not**
formalize the Lemma of `research/THM_QUOTIENT_DESCENT1.md` as stated (descent
through a degree-2 quotient), nor its Corollary. Machine status **passed** on
hosted CI for the exact bytes reviewed. Review outcome for the target's claim:
validated_scoped, with the single-reviewer and no-local-rebuild deviations
listed below. No hypothesis, ledger record or sheriff case is changed.

Bound inputs

Input	Value
Reviewed revision	`origin/main` `3fe6e1226d5a40f9f8f1e1d30c513481e34bf3a9`
Theorem	`CryptoResearch.Quotient.translation_commutes_iff_two_torsion`
Lean file	`formal/CryptoResearch/Quotient/TranslationDescent.lean`, sha256 `359a5ced…6bf401e`
Target	`formal/targets/quotient-translation-descent.yaml`, sha256 `12fa5950…bdd24b`
Source note	`research/THM_QUOTIENT_DESCENT1.md`, sha256 `ac91debe…99b08e`
Toolchain	`leanprover/lean4:v4.35.0-rc1` (`lean-toolchain` sha256 `5699bbbd…dc150`)
Mathlib	`7fa22735254ddf7e11f119250a9f2d0352bc7c3a` (`lake-manifest.json` sha256 `3adfcdc0…c3a4dc`)

The full hashes are in review.yaml.

Machine status: passed (hosted), local rebuild not possible

- **Hosted, current bytes.** `lean-verification` run
  [40760275078](https://github.com/isogeny-labs/crypto-autoresearcher/actions/runs/40760275078),
  job `formal workspace (lake build + axiom audit)` (123070123680), head
  `d75b7d47` of #2654, 2026-10-28. Plain `lake build` of the whole workspace
  (8,979 jobs, exit 0), then `tools/verify_formal_targets.py`:
  `87 verified, 0 not verified`. The receipt for this target in artifact
  `formal-verification-receipts` (id 12850205685, zip sha256
  `e2ebfcca…dbead95`, downloaded and read for this review) records
  `status: machine_verified`, `axiom_audit_passed: true`,
  `axiom audit: 700 theorem(s), all standard axioms only`, and the elaborated
  type below. Its `theorem_sha256`, `task_spec_sha256`, `lean_toolchain_sha256`
  and `lake_manifest_sha256` equal the hashes of the files on the reviewed
  `main`, so that run checked exactly the bytes reviewed here.
  Run 40713787875 (head `b0e03a16` of #2496) gives the same result.
- **`#print axioms` = `[propext]`.** The hosted gate checks that the axioms are
  standard ones; it does not print the list. The exact list `[propext]` comes
  from `docs/reports/2026-10-23-translation-commutes-rename/evidence/print-axioms.log`.
  Its receipt binds the same theorem, toolchain and manifest hashes, so the
  file has not changed since. It agrees with the proof: `abel` normalization
  and `rw` with no classical step or quotient. Treat it as replayed evidence,
  not a fresh check from this session.
- **Local rebuild: `infrastructure_error`.** This container has no Lean. Installing
  elan (`curl … | sh`) was refused by the session's permission policy, so no
  `lake build` or `#print axioms` ran here. This is not a fact about the
  theorem.

Elaborated type (from the hosted receipt):

```lean
CryptoResearch.Quotient.translation_commutes_iff_two_torsion.{u}
  {E : Type u} [AddCommGroup E] (S T : E) :
  (∀ (P : E), -(P + S) + T = -(P + T + S)) ↔ 2 • T = 0
```

Semantic fidelity

Definitions. The left-hand side spells out τ_T ∘ ι_S (P) = −(P+S) + T and ι_S ∘ τ_T (P) = −(P+T+S), with ι_S(P) = −(P+S) and τ_T(P) = P+T. Both match the note's Proof. `2 • T` is `ℕ`-smul, i.e. T + T, the note's "2T = O". Neither ι_S nor τ_T is defined as a Lean term. The identity is stated pointwise, which says the same thing.

Quantifiers. S and T are universally quantified outside. P is universally quantified inside the left side only. That is the right reading: "commutes" means equal as maps. The placement is not load-bearing here, because for each single P the two sides differ by exactly 2T. So the identity at any one P (an ∃P reading, or ∀P moved outside the ↔) is already equivalent to 2T = 0. `E` is any additive commutative group in any universe.

Hypotheses. There are none beyond `AddCommGroup`. The stub's `2 • S = 0` was dropped. That is correct: the note's 2026-10-16 correction (sheriff L2) says S may be any point, and controls M2 below show that `2S = 0` is not the condition. The note's `char k ≠ 2` assumption plays no part in the group identity and is not needed. Commutativity is used: `abel` closes the goal.

Both directions. (→) instantiates P = 0, and the difference of the two sides is 2T. (←) rewrites the left side to the right side plus 2T. Both are real directions, and neither is assumed.

Degenerate and vacuous cases. The trivial group, and any group of exponent 2, make both sides true for every S and T. That is correct and not vacuous: the controls show both sides take both truth values on 144 finite groups, false for 408,100 (S, T) pairs and true for 14,400. The condition does not depend on S at all. This is the content: no choice of involution centre admits a translation that is not 2-torsion.

Bridges not formalized (each one is prose only):

1. That $E(\bar{k})$ of an elliptic curve is the `AddCommGroup` the theorem is applied to. No elliptic curve, field, Weierstrass model or Mathlib `WeierstrassCurve.Affine.Point` appears.
2. That ι_S is an involution of the curve, i.e. a morphism with ι_S ∘ ι_S = id. This is trivial, but not stated in Lean.
3. That every degree-2 map φ: E → P^1 over k is, up to an automorphism of P^1, the quotient by some ι_S.
4. That "τ_T descends through E/⟨ι_S⟩" (there is ψ: P^1 → P^1 with φ ∘ τ_T = ψ ∘ φ) is equivalent to "τ_T commutes with ι_S". This is the fibre argument. It also needs ψ to be a morphism, not only a map of sets, which the note does not argue.
5. The Corollary's acting-group statement for summation polynomials, the bound 4^{m−1}·m!, and the three curve-class cases.

Claims supported and not supported

Claim	Where	Supported by the Lean theorem?
In any additive commutative group, for all S and T, τ_T commutes with ι_S iff 2T = 0	target `claim`; note Proof, last line	**Yes**, exactly
The identity holds for every S; `2S = 0` is unnecessary	target `notes`; note correction 2026-10-16	**Yes** (no hypothesis on S)
The same identity for the group $E(\bar{k})$ of an elliptic curve	note Setting	Only by instantiation, which is not done in Lean (bridge 1)
τ_T **descends** through φ = E/⟨ι_S⟩ iff 2T = O (the Lemma as stated)	note Lemma; title	**No**: bridges 3 and 4
Every degree-2 map E → P^1 is a quotient by some ι_S; x(P+R) is the quotient by ι_{2R}	note Setting	**No**
ι_S is an involution of the curve	note Setting	**No** (bridge 2)
The acting group on summation-polynomial solutions is E[2](k)^m ∩ {ΣT_i = O}, order ≤ 4^{m−1}·m!; (Z/4)^{m−1} needs degree > 2 or an automorphism	note Corollary	**No**
Closes the "Larger group actions?" question for every x-type coordinate	note header	**No**. This needs bridges 3–5
Novelty	note header	Not a formal question. The note itself claims only "formalized here"

The rename in #2496 (`translation_descends…` → `translation_commutes…`) removed the one mismatch between the name and the statement. The docstring, the file header ("WHAT IS NOT PROVED") and the target `notes` now describe the boundary above correctly. I found no wording on the target or in the Lean file that claims more than the theorem proves. The note's **title** and **Lemma** still state descent. The Status line and the 2026-10-17 and 2026-10-23 corrections say that only the identity is formal. Anyone citing this theorem for descent or for the Corollary would be overclaiming.

Controls

controls.py (standard library only, sha256 `7642609e…ef27b`) re-derives the statement independently of the Lean proof. It enumerates every Z/a × Z/b with a, b ≤ 12 and every S, T in each. Output: controls.out.

Control	Result
Positive: the equivalence on all 422,500 (S, T) pairs	holds everywhere
Nonvacuity: the left side takes both values	false 408,100, true 14,400
M1: conclusion `T = 0`	refuted on 8,316 pairs (non-zero 2-torsion T commutes)
M2: conclusion `2S = 0` (the stub's hypothesis)	refuted on 27,000 pairs
M3: τ_S in place of ι_S	commutes on all 408,100 pairs where 2T ≠ 0, so the [−1] carries the content

The sheriff outputs cited in the note's 2026-10-17 correction already cover curve-level checks: every T on two curves over F_1009 and F_211, with S = 2R not 2-torsion. They check the identity on curve points. They do not check bridges 3 and 4.

Deviations

- **One reviewer.** The lean-review route asks for a precommitted `review_plan`, separate reviewer sessions, one owner per joint, and a blind re-derivation by someone who has not read the producer's files. This is one reviewer, who read the Lean file and the note before writing the controls. The controls are an independent re-derivation, not a blind one, and `tools/check_review_independence.py` was not run. Because of this, the `validated_scoped` outcome covers only the target's group-law claim, where the proof is two `abel` steps. It is not a claim-changing validation of the research note.
- **No local Lean check.** See the machine-status section above. The fresh machine evidence is the hosted gate, and the exact axiom list is replayed.
- **Plain PDF.** report.pdf puts the diagram and this text into a PDF without typesetting. There is no performance chart, because this review has no performance or experiment content.

Next actions

None needed for the target. Anyone who wants the note's Lemma or Corollary formal must first state bridges 3 and 4 (degree-2 maps $E \to P^1 \cong$ quotients by ι_S, and descent $\Longleftrightarrow$ commutation). Mathlib currently has no theory of quotient curves for elliptic curves that would support this.